AQUEN FLUID TECHNOLOGIES LLP

Complete Water & Wastewater Treatment Solutions

Design | Manufacturing | Supply | Installation | Commissioning | AMC

MIXED BED
DEMINERALISATION (DI) SYSTEM

ULTRA PURE WATER FOR CRITICAL APPLICATIONS

FRP / MSEP / MSRL VESSELS | 100 LPH – 10,000 LPH

- Ultra Low Conductivity (< 1 µS/cm)
- High Purity Water
- FRP, MSEP & MSRL Options
- Automatic / Manual Operation
- Easy Regeneration & Maintenance
- Compact & Robust Design
- Suitable for Industrial Applications



FRP MIXED BED
(100 LPH – 5,000 LPH)
With Multiport Valve (MPV)



MSEP MIXED BED
(1,000 LPH – 10,000 LPH)
MS with Epoxy Lining



MSRL MIXED BED
(500 LPH – 10,000 LPH)
MS with FRP / Rubber Lining

PROCESS FLOW DIAGRAM – MIXED BED SYSTEM

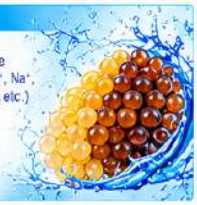


APPLICATIONS / INDUSTRIES



WORKING PRINCIPLE

Mixed bed contains a homogeneous mixture of Acid Cation Resin (SAC) in H^+ form and Strong Base Anion Resin (SBA) in OH^- form. The cations (Ca^{2+}, Mg^{2+}, Na^+, K,Fe^{2+}, etc.) and anions (Cl^-, SO_3^{2-}, NO_3^-, HCO_3^-, SiO_3 etc.) are exchanged on the resins to produce ultra pure demineralised water with very low conductivity. The resins are regenerated separately by separation and regeneration of cation and anion resins or by replacement method.



KEY FEATURES

- Produces ultra low conductivity water (< 1 µS/cm)
- Homogeneous mixture of cation & anion resin
- Available in FRP, MSEP and MSRL construction
- High exchange capacity and long resin life
- Separate regeneration of cation and anion resin
- Automatic / Manual operation options
- Compact and skid mounted design
- Low maintenance and easy resin replacement
- Suitable for continuous or intermittent operation
- Custom design as per water quality requirement

TYPICAL INLET & OUTLET PARAMETERS

Parameter	Unit	Typical Inlet (from Pre-treated Water)	Typical Outlet (After Mixed Bed)
pH	–	6.5 – 8.5	6.0 – 7.0
Total Hardness (as $CaCO_3$)	mg/L	< 1	< 0.1
Chloride (Cl^-)	mg/L	< 5	< 0.1
Sulphate (SO_4^{2-})	mg/L	< 5	< 0.1
Nitrate (NO_3^-)	mg/L	< 5	< 0.1
Silica (SiO_2)	mg/L	< 1	< 0.1
Total Dissolved Solids (TDS)	mg/L	< 10	< 1
Conductivity	µS/cm	10 – 50	< 0.1 – 1.0

OPERATING PARAMETERS

Parameter	Value
Service Flow Rate	As per table
Operating Pressure	2.5 – 6.0 kg/cm²
Maximum Pressure	6.0 kg/cm²
Operating Temperature	5 – 40 °C
Bed Depth	800 – 1,200 mm
Free Board	300 – 500 mm
Regeneration Method	Separate regeneration of cation & anion resin
Regeneration Chemical	HCl (4–10%) for Cation NaOH (4–6%) for anion
Regeneration Time	60 – 90 minutes each
Slow Rinse	30 minutes
Fast Rinse	30 minutes
Operating Bed Volume (OBV)	20 – 40 m³/hr* (depending on water quality)

TECHNICAL SPECIFICATIONS RESIN QUANTITY – MIXED BED (100 LPH TO 10,000 LPH)

Model	Flow Rate (LPH)	Vessel Type	Vessel Dia (mm)	Straight Height (mm)	Total Resin Qty (Litres)			Service Flow Rate (m³/hr)	Inlet / Outlet Size (NB)	Operating Bed Volume (m³)	Recommended OBR (m³/hr)	Resin Regeneration Qty per Cycle (kg)		Rinse Water Requirement (m³/cycle)
					Cation Resin (L)	Anion Resin (L)	Total (L)					HCl (4–10%)	NaOH (4–6%)	
AQU-MB-0.1	100	FRP	200	1200	12.5	12.5	25	0.1	15	0.025	20 – 25	0.5	0.5	0.5
AQU-MB-0.25	250	FRP	300	1400	25	25	50	0.25	20	0.05	20 – 25	1.0	1.0	1.0
AQU-MB-0.5	500	FRP	400	1650	62.5	62.5	125	0.5	25	0.125	20 – 30	2.5	2.5	1.5
AQU-MB-1.0	1,000	FRP / MSEP	500	1800	125	125	250	1.0	32	0.25	20 – 35	5.0	5.0	3.0
AQU-MB-2.0	2,000	FRP / MSEP	700	2000	200	200	400	2.0	40	0.40	20 – 35	8.0	8.0	4.5
AQU-MB-3.0	3,000	MSEP / MSRL	800	2200	300	300	600	3.0	50	0.60	20 – 35	12.0	12.0	6.0
AQU-MB-5.0	5,000	MSEP / MSRL	1,000	2400	450	450	900	5.0	65	0.90	20 – 40	18.0	18.0	8.0
AQU-MB-7.5	7,500	MSEP / MSRL	1,200	2600	600	600	1,200	7.5	80	1.20	20 – 40	24.0	24.0	10.0
AQU-MB-10	10,000	MSEP / MSRL	1,400	2800	800	800	1,600	10.0	100	1.60	20 – 40	32.0	32.0	12.0

Note: Resin quantity and regeneration chemical consumption are indicative and may vary based on raw water quality, operating conditions and resin manufacturer data. OBR: Operating Bed Volume.

TYPICAL BILL OF MATERIAL (BOM)

Sr. No.	Description	Quantity
1	Mixed Bed Vessel (FRP / MSEP / MSRL)	1 No.
2	Cation Resin (SAC)	1 Set (as per table)
3	Anion Resin (SBA)	1 Set (as per table)
4	Multiport Valve (MPV) / Butterfly Valves	1 Set
5	Piping (UPVC / CPVC / MS / SS – as required)	1 Set
6	Pressure Gauge	2 Nos.
7	Sample Valve	1 No.
8	Air Release Valve (ARV)	1 No.
9	Drain Valve	1 No.
10	Flow Meter (Optional)	1 No.
11	Control Panel (Auto / Manual)	1 No.
12	Chemical Dosing System (for regeneration)	1 Set (Optional)
13	Chemical Storage Tanks (HCl / NaOH)	2 Nos. (Optional)
14	Interconnecting Fittings, Supports & Accessories	1 Set

MATERIAL OF CONSTRUCTION (MOC)

Item	MOC Options
Vessel	FRP / MS (Epoxy Lined) / MS (FRP Lined – MSRL)
Piping	UPVC / CPVC / PP / MS / SS-304 / SS-316
Valves	UPVC / CPVC / PP / CI / SS-304 / SS-316
Distribution System	UPVC / CPVC / PP
Multiport Valve (FRP)	UPVC / ABS
Instrumentation	SS-304 / SS-316
Cation Resin	MS with Powder Coating / SS-304 (Optional)
Anion Resin	Strong Acid Cation Resin (SAC)
Gaskets / O-rings	EPDM (Food Grade) / Viton (FKM)
Supports / Skid	MS with Mirror Finish / SS-304

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CLEANER WATER GREENER TOMORROW

1. FRP MIXED BED SYSTEM – 1,000 LPH

SITE IN PROGRESS (FRP VESSEL WITH MPV)



Commercial Building Project – 1,000 LPH

FRP Mixed Bed with Multiport Valve (MPV) – Ongoing Installation

2. MSEP MIXED BED SYSTEM – 5,000 LPH

SITE IN PROGRESS (MS VESSEL WITH EPOXY LINING)



Industrial Project – 5,000 LPH

MSEP Mixed Bed with Epoxy Lining – Ongoing Installation

3. MSRL MIXED BED SYSTEM – 3,000 LPH

SITE IN PROGRESS (MS RUBBER LINED)



Power Plant Project – 3,000 LPH

MSRL Mixed Bed with Rubber Lining – Ongoing Installation

4. FRP MIXED BED SYSTEM – 500 LPH

SITE IN PROGRESS (COMPACT SKID MOUNTED)



Pharmaceutical Project – 500 LPH

FRP Mixed Bed – Skid Mounted – Ongoing Installation

5. MSEP MIXED BED SYSTEM – 7,500 LPH

SITE IN PROGRESS (MS VESSEL WITH EPOXY LINING)



Textile Industry Project – 7,500 LPH

MSEP Mixed Bed with Epoxy Lining – Ongoing Installation

6. MSRL MIXED BED SYSTEM – 10,000 LPH

SITE IN PROGRESS (MS RUBBER LINED)



Industrial Project – 10,000 LPH

MSRL Mixed Bed with Rubber Lining – Ongoing Installation

AQUEN
FLUID BY NATURE

FRP MIXED BED SYSTEM – 250 LPH

SITE IN PROGRESS – COMPACT SKID MOUNTED (SINGLE VESSEL)

ULTRA PURE WATER
FOR INDUSTRIAL APPLICATIONS

